6 Everyday use and help

FULL

11. How your website works, and how to change it safely

Quick version

No time to read it all? This is the short version. The full guide is below.

- How a visit travels through your garden, where each kind of file lives, and the safe way to change things.
- Files in `site/` (style, scripts) need no rebuild: save, then reload the page with `Ctrl+Shift+R`.
- Rust files and page templates need a rebuild: menu `M`, then `7`.
- The safe way, every time: back up (menu `6`), change one thing, rebuild if needed, test (menu `5`), then keep it or undo it.
- `git diff` shows exactly what you changed. `git checkout -- FILE` undoes it.
- Settings and passwords are in `.env`: change them with menu `8`. Never paste `.env` or a password into Gemini.

What you will learn

How a visit travels through your garden, where each kind of file lives, and the safe way to change things: back up, change one thing, test, keep it (or undo).

How a visit travels

```
 browser --> Caddy (front door, ports 80/443)
               |-- /auth ...   --> sign-in page (Authelia)         container
garden-auth
               |-- /beds ...   --> garden tools (Python)            container
junsgarden-ingest
               |                   only after signing in
               |-- everything else --> the website (Rust)           container
garden-site
                                     |-- asks the database         container
garden-db
                                     |-- reads files in site/      (style,
scripts, app download)
                                     |-- reads photos in local/images
```

Before the front door is set up, you visit the website directly at `http://localhost:8030` on the garden computer.

Where things live

Folder or file	What is in it	Change it how
`site/`	Style (`style.css`), scripts (`js/`), page cards (`sections/`), the app download (`apps/`)	Edit, save, reload the page with `Ctrl+Shift+R`. No rebuild needed.
`rust/junsgarden/templates/`	The HTML of each page. `{{ ... }}` are the holes Rust fills in.	Edit, then rebuild (menu `M`, `7`)
`rust/junsgarden/src/pages/`	One Rust file per page. `catalog.rs` makes `/catalog.php`.	Edit, then rebuild

Folder or file	What is in it	Change it how
`rust/junsgarden/src/main.rs`	The list of web addresses and which function answers each	Edit, then rebuild
`local/images/`	Your photos (in `Database/`, `Gallery/`, `post/`, `blog/`, `plan/`, `HeroBanner/`)	Upload through the website. Do not rename them by hand: to rename, use menu `M`, then `18` (guide 26).
`local/data/`	Comments, guestbook, the facts on the home page	Through the website
`.env`	Settings and passwords	Menu `8` Settings. You never need to open it.
`docker-compose.yml`	How every program starts (every line has a comment)	Read it to learn. Change it with care.

The Rust code has comments on almost every line, written for learning. `docs/rust-port/CODE-TOUR.md` walks through it in order.

The safe way to change anything (every time)

1. **Back up:** menu `6`, then `1`. It takes a minute.
2. **Change one thing.** Just one.
3. **Rebuild if it was a Rust or template file:** menu `M`, then `7`. If the build fails, the red lines name the file and the line. That is Rust checking your work.
4. **Test:** open the page, and run the checklist (menu `5`).
5. **Keep it, or undo it.** In a terminal in the garden folder:

```
git diff                        # shows exactly what you changed
git checkout -- FILE            # undoes your change to FILE
git add -A && git commit -m "Changed the catalogue heading"   # keeps it
```

Example 1: change a colour (no rebuild)

1. Open `site/style.css` in the Text Editor (right-click it, then "Open With Text Editor").
2. Search (`Ctrl+F`) for a colour such as `#2f7d32` and change it.
3. Save, then reload the website with `Ctrl+Shift+R`.

Example 2: change words on a page (rebuild)

1. Find the page's template, for example `rust/junsgarden/templates/catalog.html`.
2. Change a heading's text and save.
3. Menu `M`, then `7` (rebuild). Wait for the health line `{"status":"ok", ...}`.
4. Reload the page.

Run the tests (optional)

`bash garden.sh test` runs the 64 automatic checks that come with the code. The first time downloads Rust (about 1.5 GB) and takes a while. After that it is quick.

Working with Gemini on the code

- Gemini CLI reads `GEMINI.md` in this folder, which holds the rules for helping you (guide 08).
- Paste the exact red error lines, and say which file you changed.
- Never paste `.env` or a password.
- Check what it suggests with `git diff` before you keep it.

Read more (official)

- Rust, the book: https://doc.rust-lang.org/book/

- axum (the web server part): https://docs.rs/axum
- askama (the page templates): https://askama.readthedocs.io/
- sqlx (talking to the database): https://docs.rs/sqlx
- git basics: https://git-scm.com/book/en/v2/Git-Basics-Undoing-Things